1.2 A Catalog of Essential Functions

Calculus 1

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Textbook: James Stewart, Essential Calculus - Early Transcendentals

Outline

< Part 1 >

Mathematical Modeling

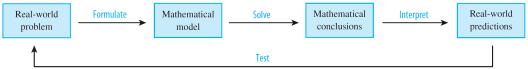
Mathematical Modeling

A **mathematical model** is a mathematical description of a real-world phenomenon such as the size of a population, the demand for a product, the speed of a falling object.

The purpose of the model is to understand the phenomenon and perhaps to make predictions about future behavior.

Process of mathematical modeling

The modeling process



Linear model

A mathematical model is never a completely accurate representation of a physical situation—it is an idealization.

Linear Models

When we say that y is a **linear function** of x, we mean that the graph of the function is a line, so, we can use the equation of a line as

$$y = mx + b$$

where m is the slope of the line and b is the y-intercept.

A characteristic feature of linear functions is that they grow at a constant rate.

Example 1

(a) As dry air moves upward, it expands and cools. If the ground temperature is $20^\circ C$ and the temperature at a height of $1km$ is $10^\circ C$, express the temperature T(in $^\circ C$) as a function of the height h (in kilometers), assuming that a linear model is appropriate.

(b) Draw the graph of the function in part (a). What does the slope represent?

(c) What is the temperature at a height of $2.5km$?

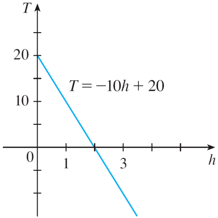
Example 1 - Solution

(a) We have $T = mh + b$.

- We are given that $T = 20$ when $h = 0$, so $20 = m \cdot 0 + b = b$.
- In other words, the y-intercept is $b = 20$.
- We are also given that $T = 10$ when $h = 1$, so the graph passes through $(1, 10)$.
- The slope of the line is therefore $m = \dfrac{10 - 20}{1 - 0} = -10$.
- Therefore, $T = -10h + 20$.

Example 1 - Solution

(b) The graph: The slope is $m = -10^\circ C/km$. This represents the rate of change of temperature with respect to height.



(c) At a height of $h = 2.5km$, the temperature is

$$T = -10(2.5) + 20 = -5^\circ C.$$

Note

< Part 2 >

Polynomial and rational functions

Polynomial

A function P is called a **polynomial** if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where n is a nonnegative integer and $a_0, a_1, a_2, \ldots, a_n$ are constants called the **coefficients** of the polynomial.

If the leading coefficient $a_n \neq 0$, then the degree of the polynomial is n.

Linear, quadratic, cubic functions

A polynomial of degree 1 is of the form $P(x) = mx + b$ and so it is a **linear function**.

A polynomial of degree 2 is of the form $P(x) = ax^2 + bx + c$ and is called a **quadratic function**.

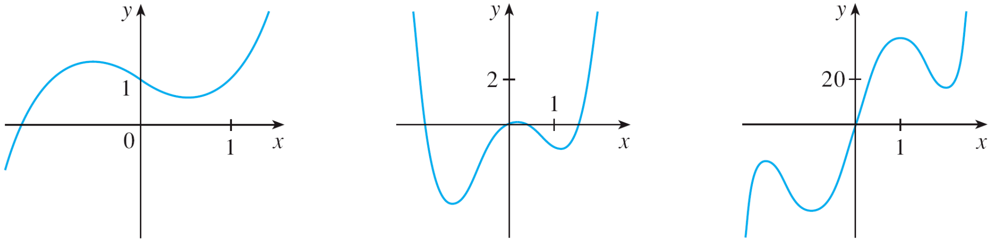
A polynomial of degree 3 is of the form $P(x) = ax^3 + bx^2 + cx + d$ and is called a **cubic function**.

Graph

(a) $y = x^3 - x + 1$

(b) $y = x^4 - 3x^2 + x$

(c) $y = 3x^5 - 25x^3 + 60x$



Power functions

A function of the form $f(x) = x^a$, where a is a constant, is called a **power function**.

We consider several cases:

(i) $a = n$, where n is a positive integer

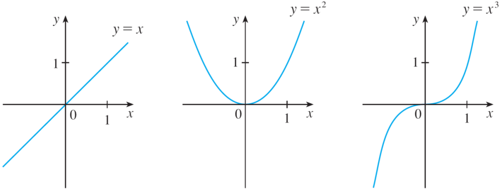
(ii) $a = 1/n$, where n is a positive integer

(iii) $a = -1$

Case 1

(i) $a = n$**, where** n **is a positive integer**

- The graphs of $f(x) = x^n$ for $n = 1, 2, 3$

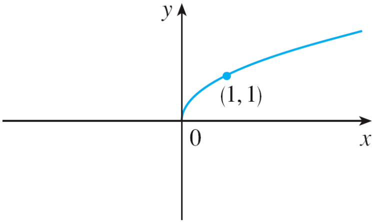


- The general shape of the graph of $f(x) = x^n$ depends on whether n is even or odd.
- If n is even, then $f(x) = x^n$ is even and its graph is similar to $f(x) = x^2$.
- If n is odd, then $f(x) = x^n$ is odd and its graph is similar to $f(x) = x^3$.

Case 2

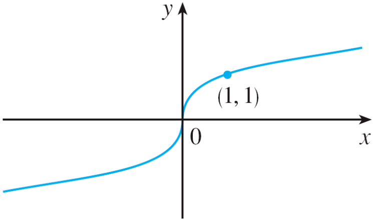
(ii) $a = 1/n$, where n is a positive integer

- The function $f(x) = x^{1/n} = \sqrt[n]{x}$ is a **root function**.
- For $n = 2$ it is the square root function $f(x) = \sqrt{x}$
- For other **even** values of n, the graph of $y = \sqrt[n]{x}$ is similar to that of $y = \sqrt{x}$.



Case 2

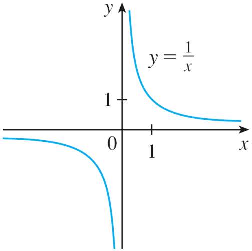
- For $n = 3$ we have the cube root function $f(x) = \sqrt[3]{x}$
- The graph of $y = \sqrt[n]{x}$ for n odd $(n > 3)$ is similar to that of $y = \sqrt[3]{x}$.



Case 3

(iii) $a = -1$

- The graph of the **reciprocal function** $f(x) = x^{-1} = 1/x$:



Rational function

Its graph has the equation $y = 1/x$, and is a hyperbola with the coordinate axes as its asymptotes.

A **rational function** f is a ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials.

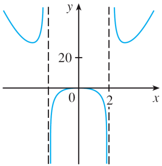
The domain consists of all values of x such that $Q(x) \neq 0$.

Example

The function

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

is a rational function with domain $\{x \mid x \neq \pm 2\}$.



Note

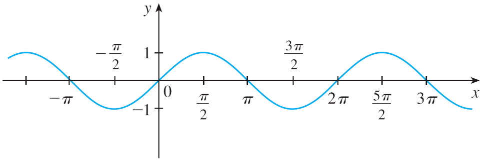
< Part 3 >

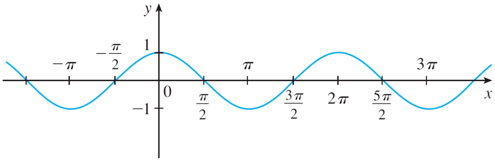
Trigonometric, exponential, logarithm functions

Sine and cosine functions

Trigonometric Functions

The graphs of the **sine and cosine functions**





Property

$$-1 \leq \sin x \leq 1, \qquad -1 \leq \cos x \leq 1$$

$$|\sin x| \leq 1, \qquad |\cos x| \leq 1$$

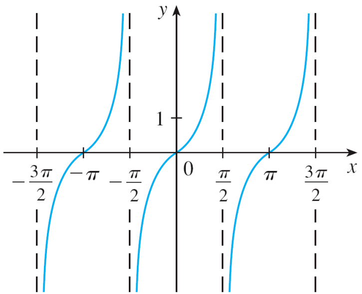
$$\sin x = 0 \qquad \text{when} \;\; x = n\pi, \;\; n \text{ an integer}$$

$$\sin(x + 2\pi) = \sin x, \qquad \cos(x + 2\pi) = \cos x$$

Tangent function

The tangent function is related to the sine and cosine functions by the equation

$$\tan x = \frac{\sin x}{\cos x}$$



Tangent function

It is undefined whenever $\cos x = 0$, that is, when

$$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$$

Its range is $(-\infty, \infty)$.

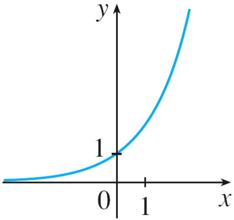
Notice that the tangent function has period π:

$$\tan(x + \pi) = \tan x$$

Exponential function

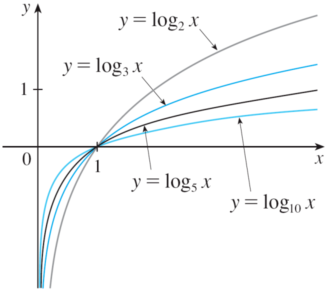
The **exponential functions** are the functions of the form $f(x) = a^x$, where the base a is a positive constant.

The graph of $y = 2^x$:



Logarithmic function

The **logarithmic functions** $f(x) = \log_a x$, where the base a is a positive constant, are the inverse functions of the exponential functions.



Note

< Part 4 >

Transformations of Functions

Translations

By applying certain transformations to the graph of a given function we can obtain the graphs of certain related functions.

Translations (or shifts) Vertical and Horizontal shift.

VERTICAL AND HORIZONTAL SHIFTS Suppose $c > 0$. To obtain the graph of

$y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward

$y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward

$y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right

$y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left

Stretching and reflecting

Stretching and **reflecting** transformations.

VERTICAL AND HORIZONTAL STRETCHING, SHRINKING, AND REFLECTING

Suppose $c > 1$. To obtain the graph of

$y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c

$y = (1/c)f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c

$y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of c

$y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of c

$y = -f(x)$, reflect the graph of $y = f(x)$ about the x-axis

$y = f(-x)$, reflect the graph of $y = f(x)$ about the y-axis

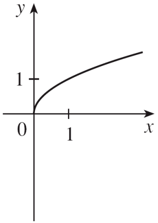
Example 2

Given the graph of $y = \sqrt{x}$ use transformations to graph

$$y = \sqrt{x} - 2 \quad y = \sqrt{x-2} \quad y = -\sqrt{x} \quad y = 2\sqrt{x} \quad y = \sqrt{-x}$$

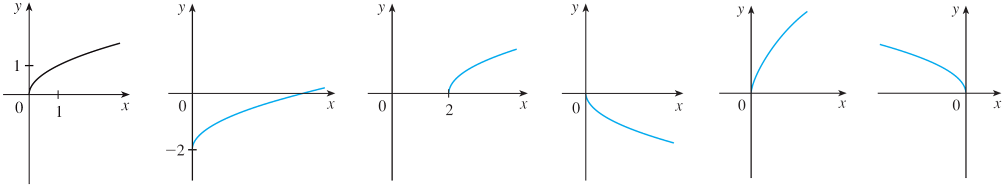
Solution:

The square root function $y = \sqrt{x}$:



Example 2 - Solution

$$y = \sqrt{x} - 2 \quad y = \sqrt{x-2} \quad y = -\sqrt{x} \quad y = 2\sqrt{x} \quad y = \sqrt{-x}$$



Note

< Part 5 >

Combinations of Functions

Sum and different functions

The sum and difference functions are defined by

$$(f+g)(x) = f(x) + g(x), \qquad (f-g)(x) = f(x) - g(x)$$

If the domain of f is A and the domain of g is B, then the domain of $f+g$ is the intersection $A \cap B$ because both $f(x)$ and $g(x)$ have to be defined.

Product and quotient functions

The product and quotient functions are defined by

$$(fg)(x) = f(x)g(x), \qquad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

The domain of fg is $A \cap B$,

but we can't divide by 0 and so the domain of f/g is

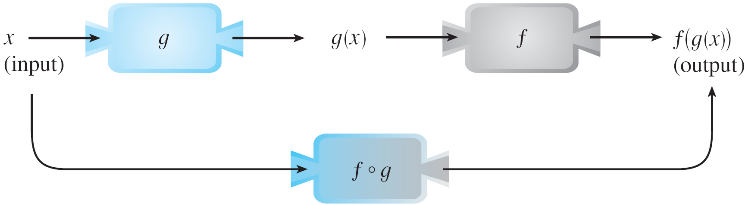
$$\{x \in A \cap B \mid g(x) \neq 0\}.$$

Composite function

A new function $h(x) = f(g(x))$ is called the **composition** (or **composite**) of f and g and is denoted by $f \circ g$ ("f circle g").

$(f \circ g)(x)$ is defined whenever both $g(x)$ and $f(g(x))$ are defined.

Figure of composite function



Example 5

If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each function and its domain.

(a) $f \circ g$ **(b)** $g \circ f$ **(c)** $f \circ f$ **(d)** $g \circ g$

Solution:

(a) $(f \circ g)(x) = f(g(x)) = f\left(\sqrt{2-x}\right) = \sqrt{\sqrt{2-x}} = (2-x)^{1/4}$

- **Domain** of $f \circ g$ is $(-\infty, 2]$.

(b) $(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2-\sqrt{x}}$

- **Domain:** For $\sqrt{x}$ to be defined we must have $x \geq 0$.
- For $\sqrt{2-\sqrt{x}}$ to be defined we must have $2-\sqrt{x} \geq 0$, that is, $\sqrt{x} \leq 2$, or $x \leq 4$.
- Thus, we have $0 \leq x \leq 4$, so the domain of $g \circ f$ is $[0, 4]$.

Example 5 - Solution

(c) $(f \circ f)(x) = f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = x^{1/4}$

- The domain of $f \circ f$ is $[0, \infty)$.

(d) $(g \circ g)(x) = g(g(x)) = g\left(\sqrt{2-x}\right) = \sqrt{2 - \sqrt{2-x}}$

- This expression is defined when both $2 - x \geq 0$ and $2 - \sqrt{2-x} \geq 0$.
- First: $x \leq 2$.
- Second: $\sqrt{2-x} \leq 2$, or $2 - x \leq 4$, or $x \geq -2$.
- Thus, $-2 \leq x \leq 2$, so the domain of $g \circ g$ is $[-2, 2]$.

More compositions

It is possible to take the composition of three or more functions.

For instance, the composite function $f \circ g \circ h$ is found by first applying h, then g, and then f as follows:

$$(f \circ g \circ h)(x) = f(g(h(x))).$$

Note